

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1376

Switch 138 kV
July 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1376. The Interconnection Customer proposes a 288.5 MW solar photovoltaic facility interconnecting at the Switch 138 kV substation in the Pennsylvania area. The request was received on January 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,254,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Randolf Ctr – Switch 138 kV	Middletown Substation – Monaco 138 kV	167.2	187	47.9
Reedville – Randolf Ctr 138 kV	Middletown – Monaco 138 kV line	160.1	181	47.9
Bpu Solar Farm – Switch 138 kV line	Randolf Ctr – Switch 138 kV	127.5	207	100.0
Middletown – Monaco 138 kV	Randolf Center – Switch 138 kV L	125.3	248	42.2
Marcus Hook Energy Lp – Middletown 138 kV line	Switch – Randolf Ctr 138 kV L	119.7	252	42.2
Red Rock – Lafayette Catepillar 138 kV	Chichester – Lake Cowichan 138 kV	119.2	233	23.8
Reedville – Lafayette Catepillar 138 kV line	Lake Cowichan – Chichester 138 kV Line	115.9	259	23.8
Bpu Solar Farm – Monaco 138 kV	Randolf Ctr – Switch 138 kV	110.8	220	100.0
Liberty Electric Power Plant – Warehouse Electric 138 kV L	Sea Island – 6Th Street 138 kV	101.5	366	7.9
Green Counry – Red Rock 138 kV line	Bridgeport Substation – Lake Catherine 138 kV Line	100.8	369	18.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Randolf Ctr – Switch 138 kV	Rebuild line with higher-capacity conductor	\$3,185,000	60% (\$1,901,000)
Reedville – Randolph Center 138 kV	Rebuild line with higher-capacity conductor	\$4,681,000	60% (\$2,794,000)
Bpu Solar Farm – Switch 138 kV line	Rebuild line with higher-capacity conductor	\$829,000	100% (\$829,000)
Middletown – Monaco 138 kV	Rebuild line with higher-capacity conductor	\$1,766,000	43% (\$763,000)
Marcus Hook Energy Lp – Middletown 138 kV	Rebuild line with higher-capacity conductor	\$6,771,000	43% (\$2,923,000)
Red Rock – Lafayette Catepillar 138 kV	Rebuild line with higher-capacity conductor	\$957,000	70% (\$671,000)
Reedville – Lafayette Catepillar 138 kV	Rebuild line with higher-capacity conductor	\$2,319,000	70% (\$1,628,000)
Bpu Solar Farm – Monaco 138 kV	Rebuild line with higher-capacity conductor	\$3,277,000	100% (\$3,277,000)
Liberty Electric Power Plant – Warehouse Electric 138 kV	Reconductor line	\$485,000	42% (\$202,000)
Green Counry – Red Rock 138 kV line	Reconductor line	\$464,000	57% (\$266,000)
	Total allocated to project		\$15,254,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.